

Road Damage Detection & Severity Estimation

Transfer learning with YOLO26 on the RDD2022 China-Drone dataset

Team

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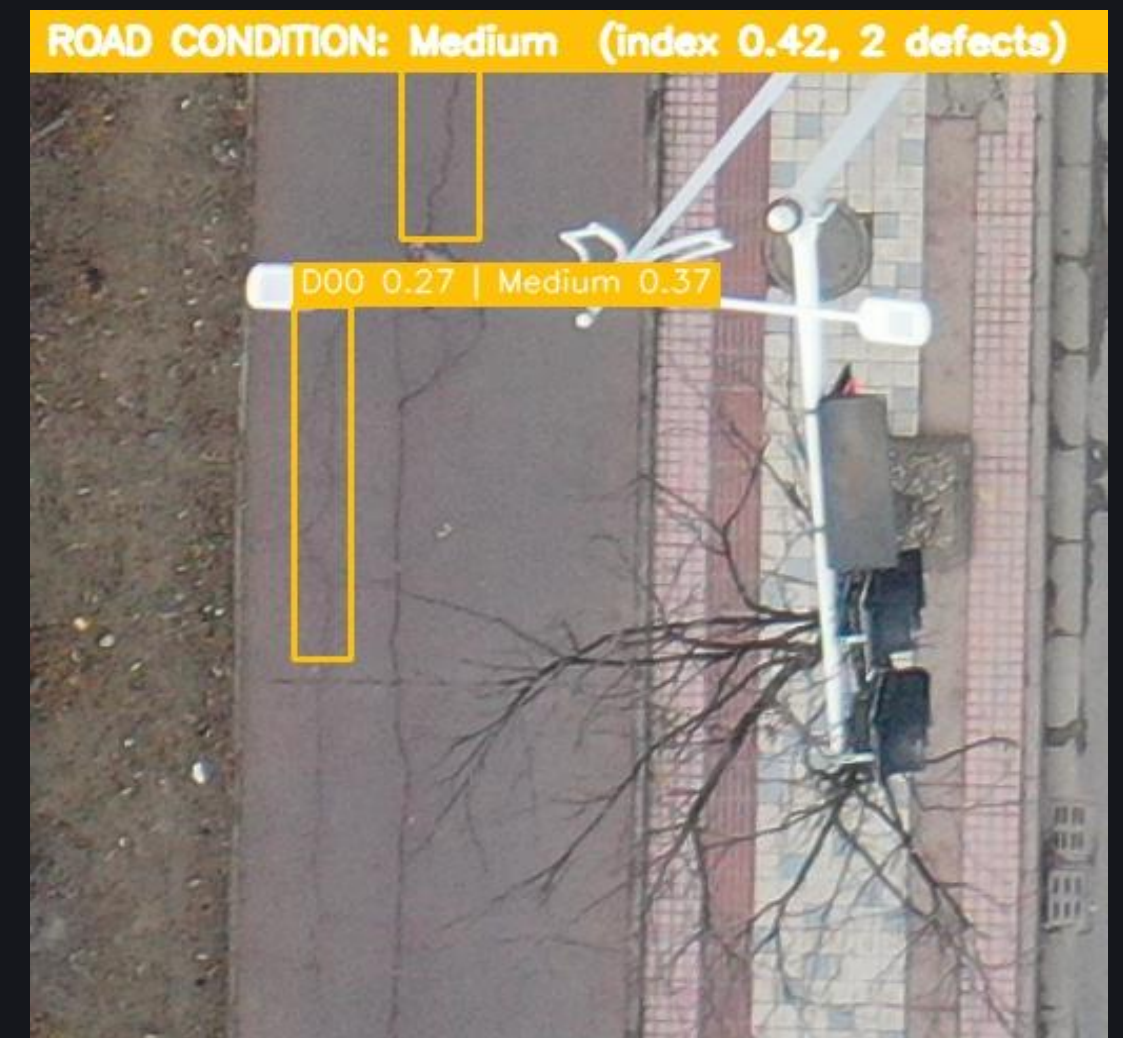
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Computer Vision Group Assignment



Why automated road inspection matters

SAFETY

Potholes are a direct hazard

Unrepaired defects cause crashes, vehicle damage and injury.

COST

Early repair is **10×** cheaper

Sealing a hairline crack beats resurfacing a failed road.

SCALABILITY

Imagery already exists at scale

Drones and dashcams produce road footage continuously.

OUR GOAL Detect & classify damage, triage severity into **Low** / **Medium** / **High**.



RDD2022 China-Drone subset

300

drone images: 512×512 px

514

labelled defect boxes

1.71

avg. defects per image

4

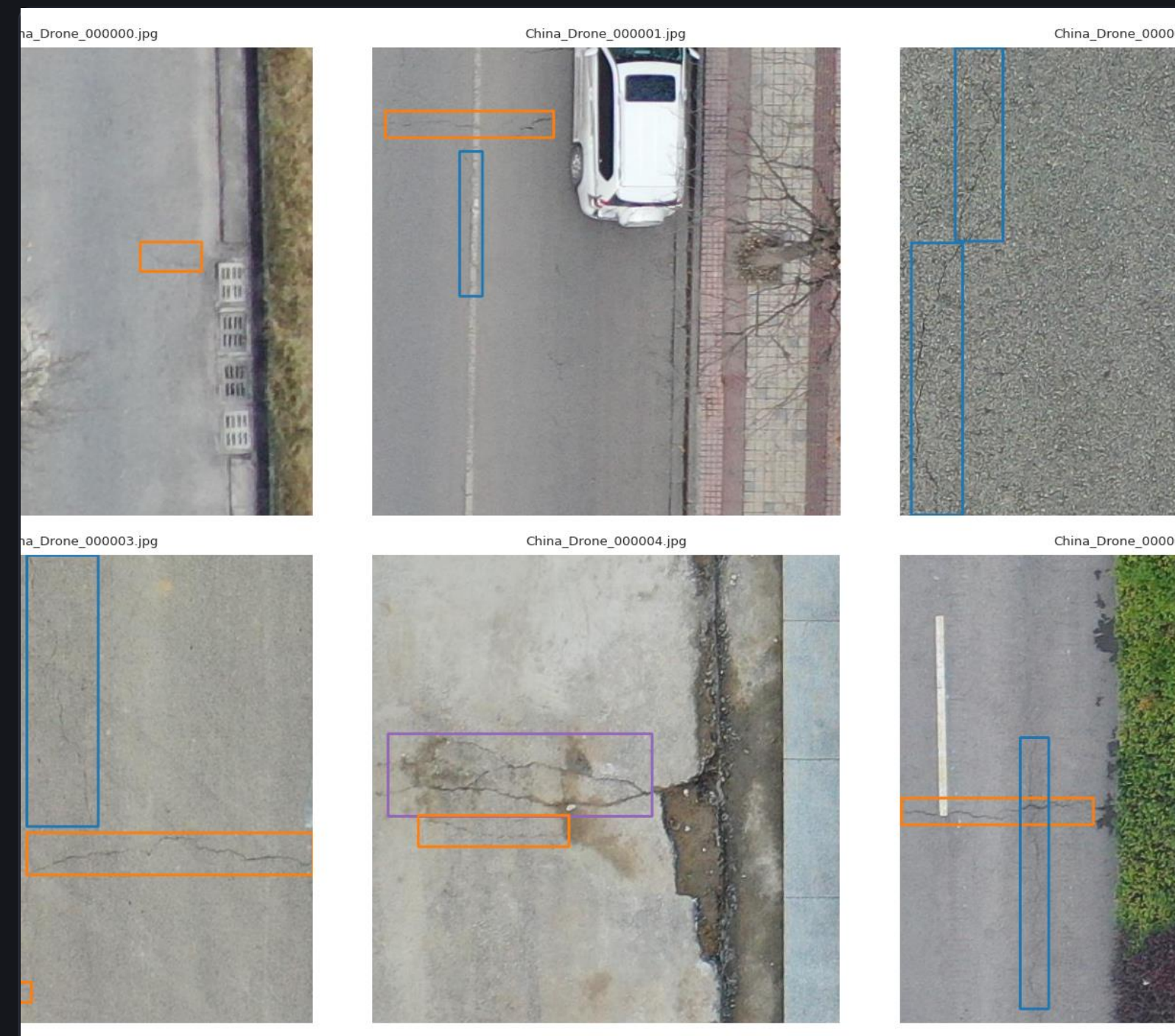
damage classes

D00 Longitudinal crack

D10 Transverse crack

D20 Alligator crack

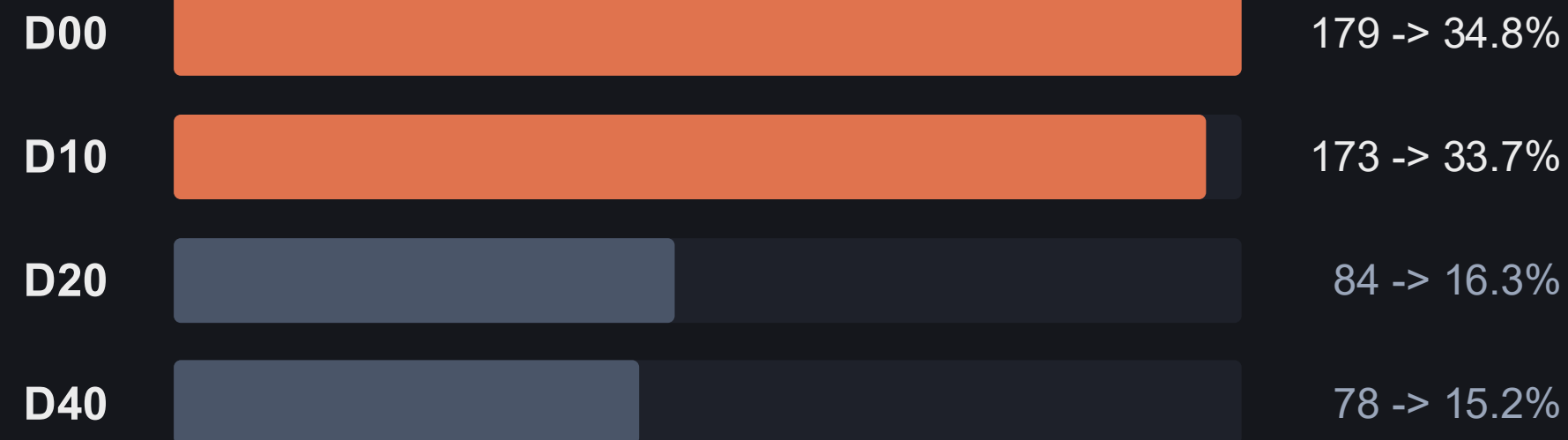
D40 Pothole



■ Longitudinal (D00) ■ Transverse (D10) ■ Alligator (D20) ■ Pothole (D40)

Source: RDD2022 (CRDDC'2022) · CC BY 4.0

Class distribution is uneven



Class imbalance

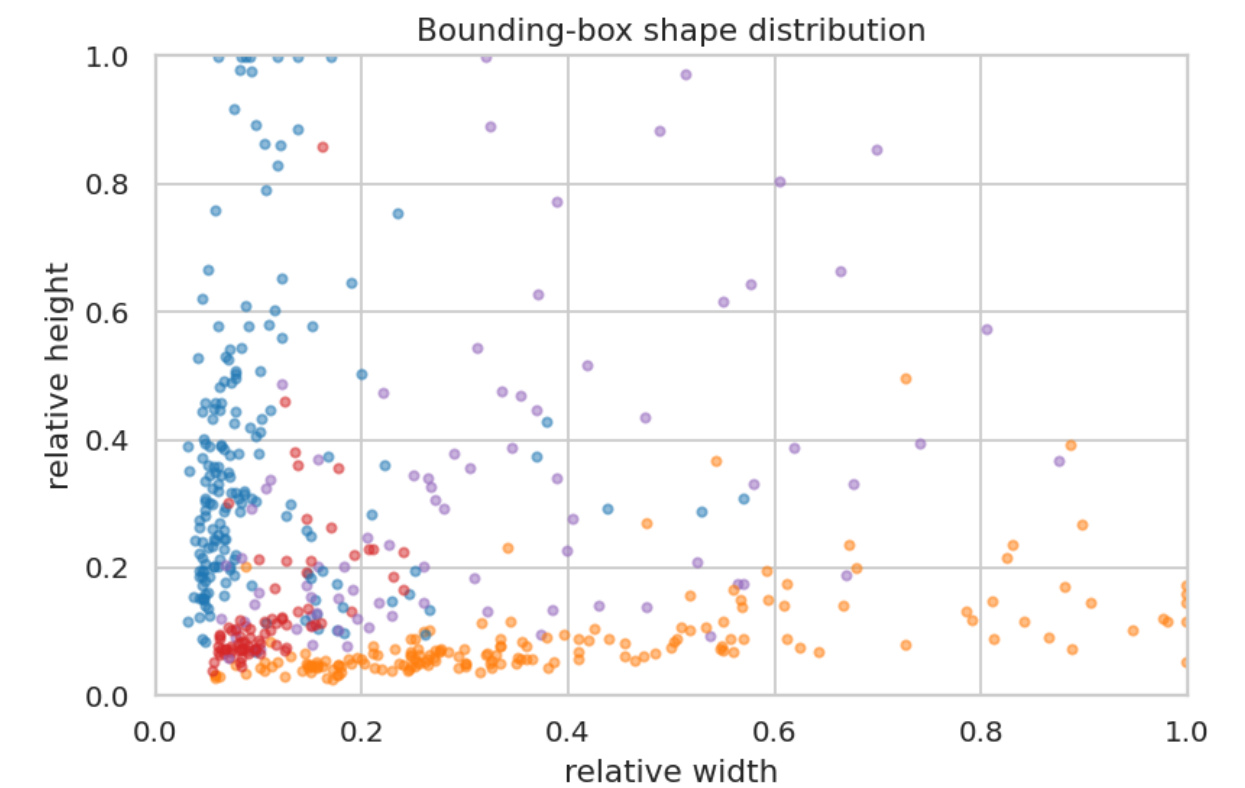
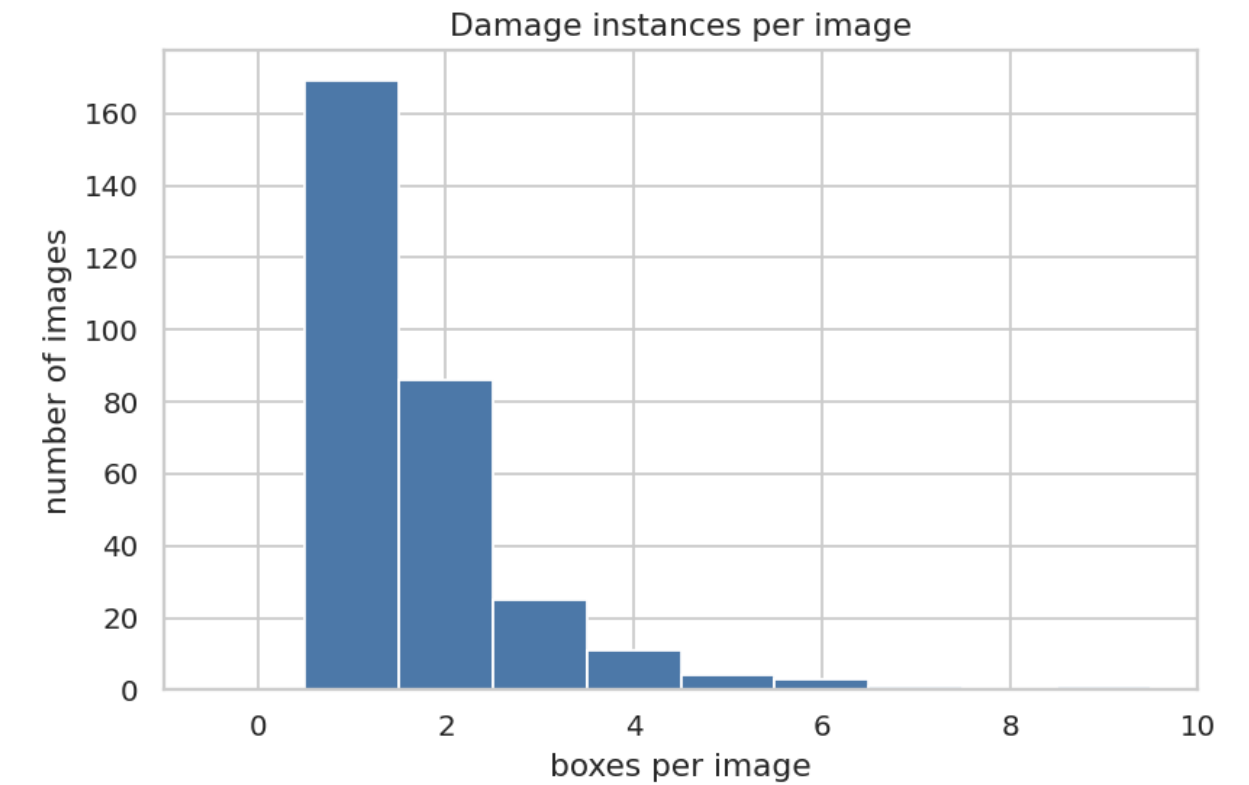
D00 + D10 = **68%** of boxes.

Small objects

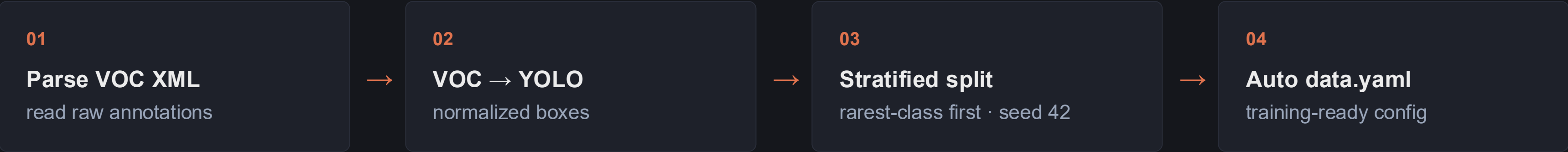
Extreme aspect ratios.

No background

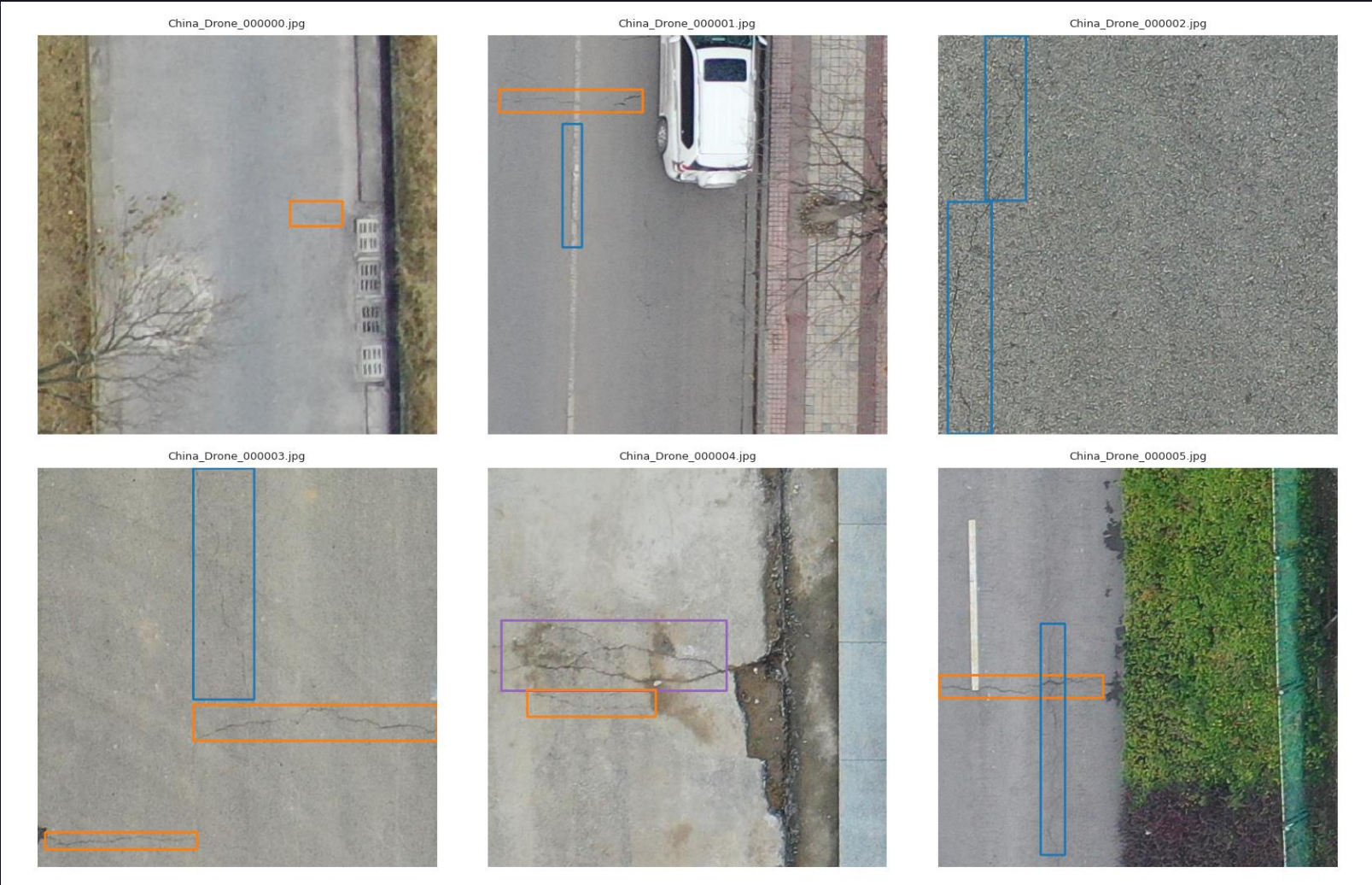
Every image has defects.



A reproducible VOC → YOLO pipeline

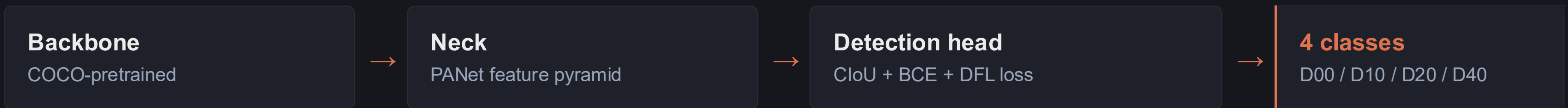


SPLIT	IMAGES	D00	D10	D20	D40
Train	210	134	113	60	54
Val	60	29	43	16	17
Test	30	16	17	8	7



Annotated samples: YOLO normalised boxes ready for training

YOLO26 with transfer learning



WHY TRANSFER LEARNING

Only 210 training images which is far too few to train from scratch. COCO-pretrained weights provide generic visual primitives (edges, textures, shapes) that transfer directly to road imagery. Fine-tuning adapts the higher layers and detection head to the 4 road-damage classes. Result: fast convergence, much less overfitting.

YOLO26n
2.5M

params: 63.9 FPS

YOLO26s
9.9M

params: 62.1 FPS

★ BEST

YOLO26m
21.8M

params: 49.3 FPS

Hyperparameters & schedule

Optimizer AdamW

lr0 1e-3 → cosine

Epochs 100 · patience 20

Batch 16

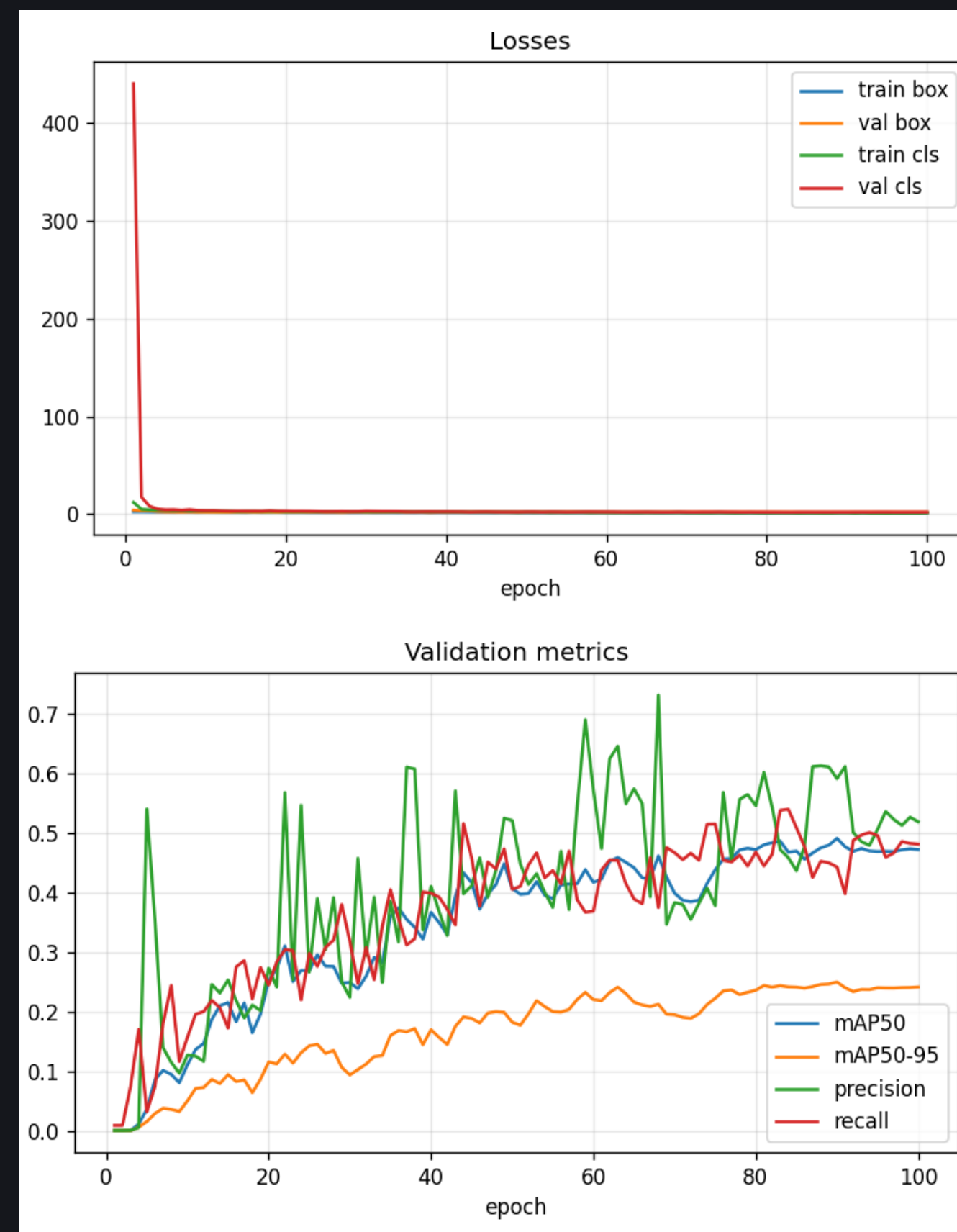
Image size 512×512

Loss CIoU + BCE + DFL

Augmentation Mosaic · HSV · flip · scale

Hardware Colab Tesla T4

Mosaic closed for last 10 epochs: fine-tuning on natural images.



YOLO26s training curve: losses converge; validation metrics climb across 100 epochs

YOLO26s wins on the balanced metrics

0.540

mAP@50

YOLO26s

0.521

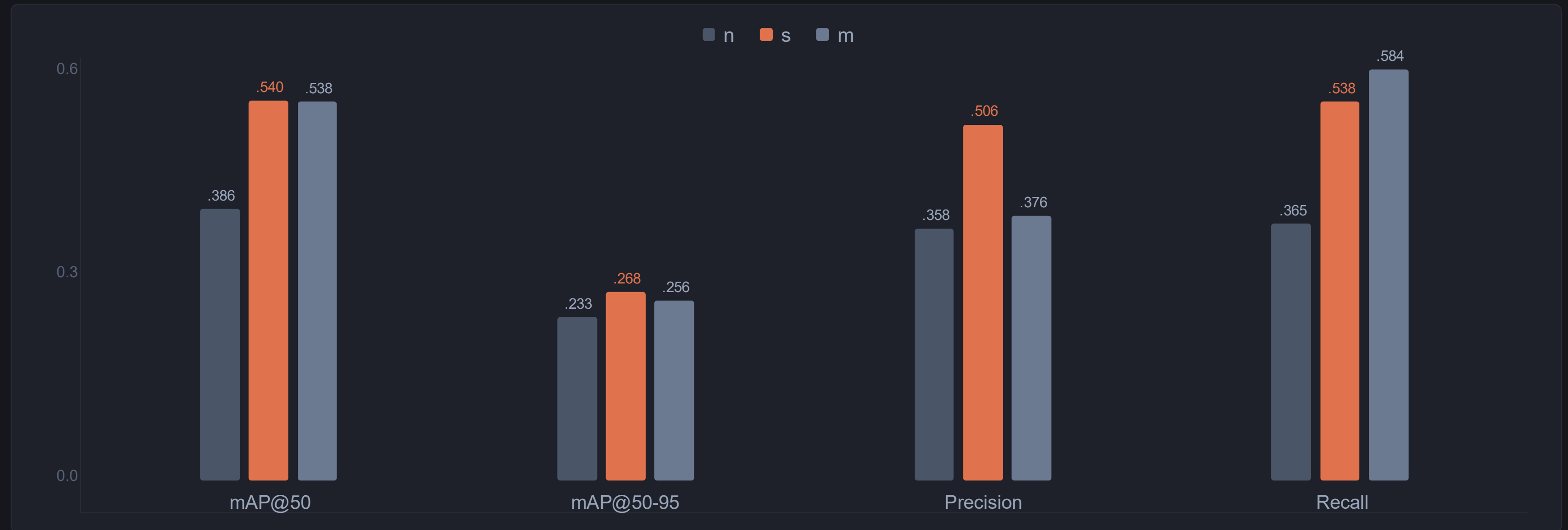
F1 score

YOLO26s

0.584

Recall

YOLO26m ✓



Key finding YOLO26s wins on accuracy, but YOLO26m has the best recall (0.584), so we **deploy YOLO26m**: triage punishes misses, not false alarms.

Potholes detect best, alligator worst

D40 Pothole ★ 0.875

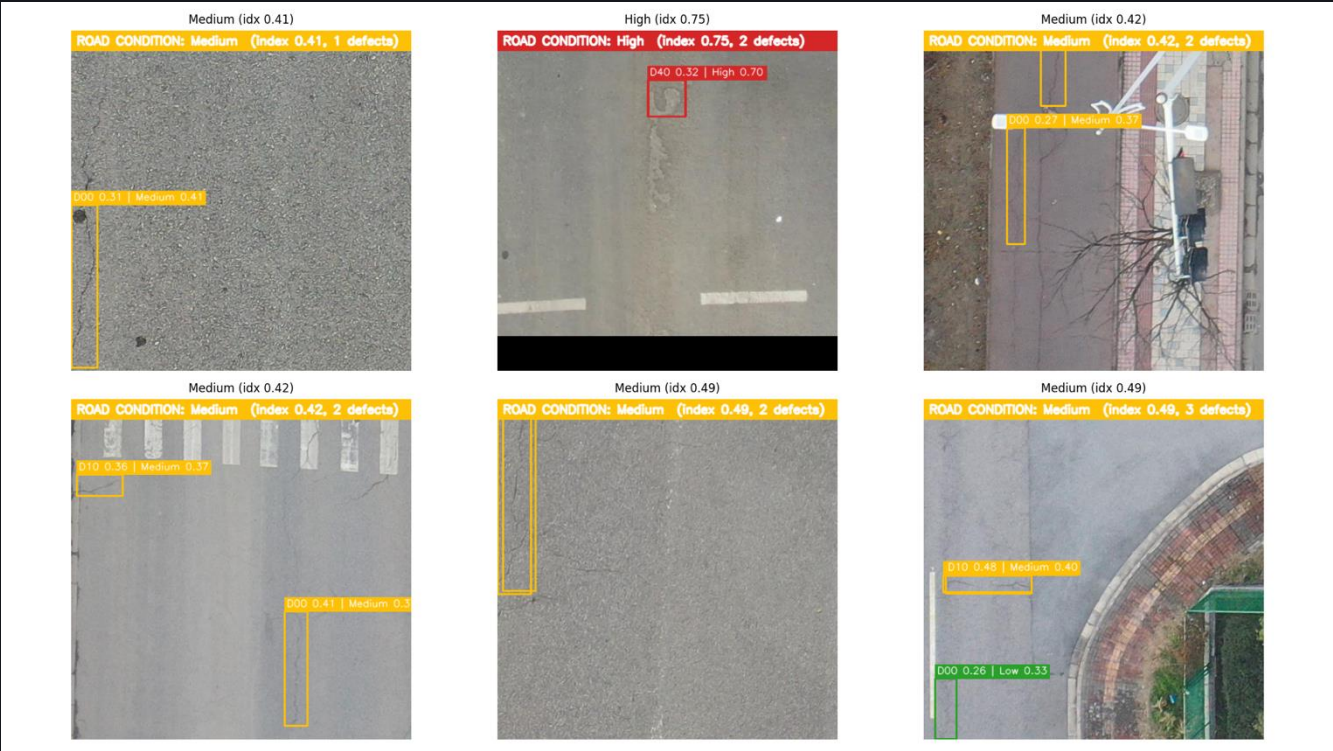
recall 1.00

D10 Transverse crack 0.611

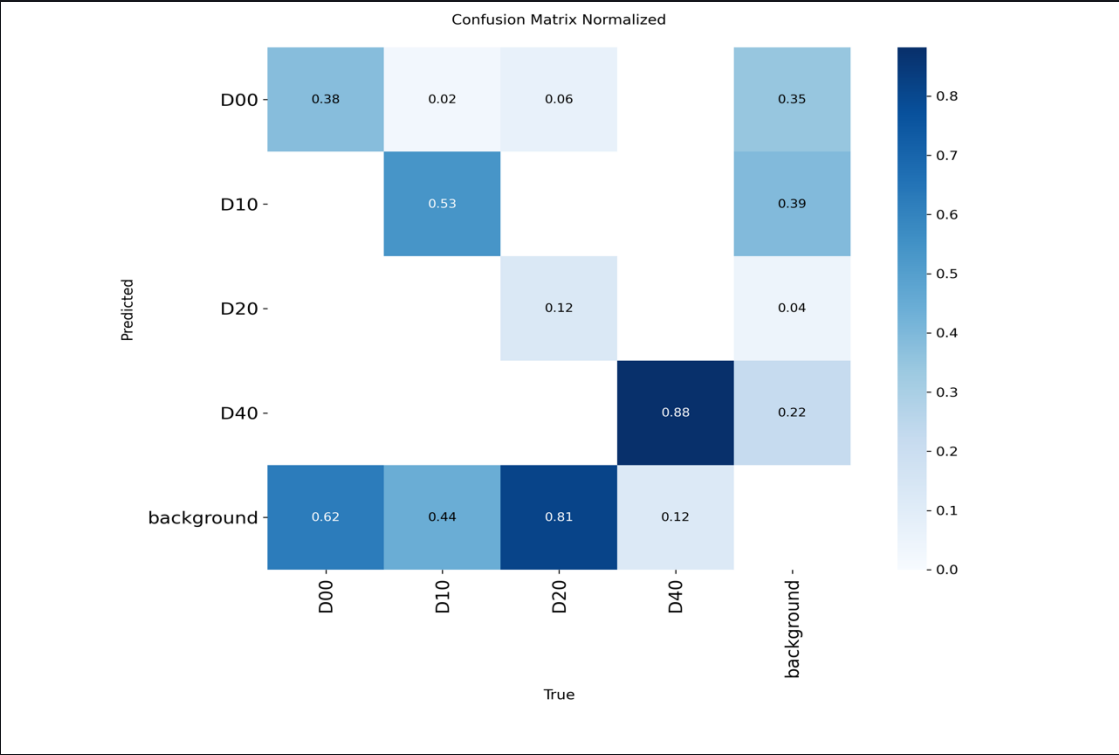
D00 Longitudinal crack 0.487

D20 Alligator crack ↓ 0.186

recall 0.125



Qualitative predictions · per-defect severity



Confusion matrix (normalised)

SURPRISE Potholes (the rarest class) score best: dark, compact blobs. Alligator is hardest: diffuse texture blends into road surface.

A transparent severity index

$$\text{severity} = \text{clip}(\text{base[class]} + \text{geometry_term}, 0, 1) \rightarrow \text{Low} / \text{Medium} / \text{High}$$

BASE SCORES BY CLASS

Pothole	0.60	Alligator	0.55
Transverse	0.30	Longitudinal	0.25

GEOMETRY TERM

Potholes & alligator scale with $\sqrt{\text{area}}$; linear cracks with $\text{length} \times \text{thickness}$.

IMAGE INDEX

$\text{max_severity} + 0.05 \cdot (\text{n_defects} - 1)$

Pothole0.723HIGH

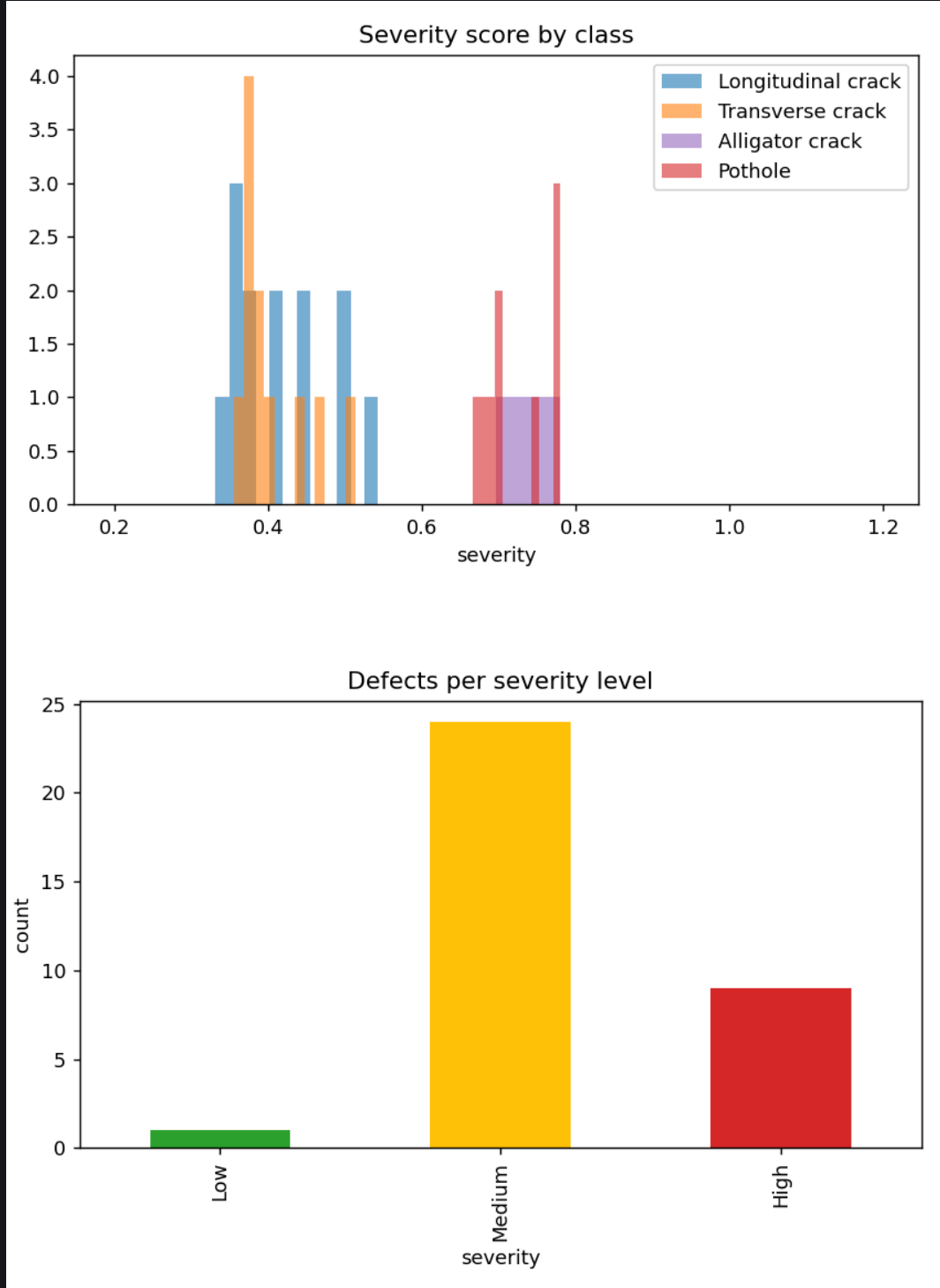
Alligator0.696HIGH

Cracks~0.41MED

Low1

Medium24

High9



| Live Demo

Gradio Dashboard

Drop in a road image: get detection, severity and a condition index live.

- 01 Upload a new road image
- 02 Detection boxes per defect
- 03 Per-defect severity in **traffic-light** colors
- 04 Overall road-condition index
- 05 Adjustable confidence-threshold slider

[Dashboard Link](#)



Failure modes & targeted fixes

Missed faint cracks
dominant failure

→ SAHI sliced / high-res inference

Alligator under-detected
recall 0.125

→ Add a segmentation head

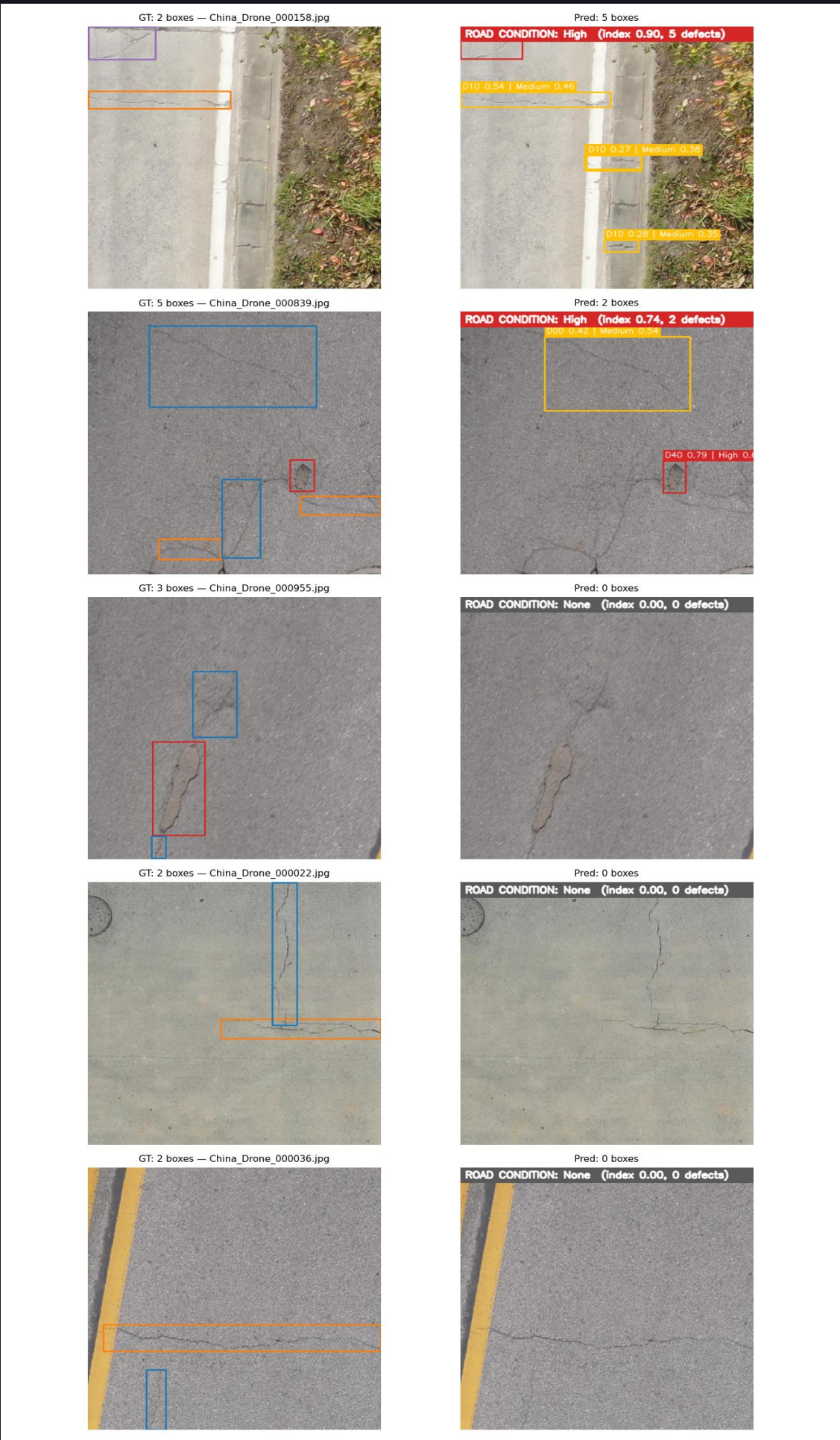
Over-counted / fragmented
split single crack

→ Merge collinear boxes

D00 ↔ D10 confusion
crack orientation

→ Adjust augmentation

BOTTOM LINE 4 of 5 worst cases are misses. Recall is the binding constraint.
Missing a pothole is costlier than a spurious box.



Worst cases: ground truth (left) vs prediction (right)

What's next

LIMITATIONS

- Small dataset: Only 210 training images vs real-world variance
- No severity ground truth to validate against
- Scale ambiguity: No depth information
- Alligator class still poorly detected

ROADMAP

- Instance segmentation (YOLO-seg / Mask R-CNN) for alligator
- SAHI sliced inference for faint cracks
- Learn severity by regression vs expert / repair-cost data
- Geo-tag detections → maintenance heat-map

Deployment choice is YOLO26m: for triage, recall matters more than precision.

| Key Takeaways

- End-to-end pipeline: VOC → YOLO → detection → severity → live demo
- YOLO26s most accurate (mAP@50 0.54), yet we deploy YOLO26m for higher recall
- Potholes easiest (0.875), alligator hardest (0.186)
- Transparent severity index passes face-validity
- Recall is the binding constraint for safety-critical triage

Thank you.

Questions?